IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired

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***Abstract*—Indoor navigation and situational awareness pose significant challenges to people with visual impairment due to the limited utility of GPS indoors and conventional mobility aids. This paper presents an IoT-enabled smart spectacle solution for improved indoor mobility and safety. This approach employs a modular edge-cloud architecture. Its physical layer is implemented on a Raspberry Pi device to support edge processing and consists of an RGB camera, LiDAR distance sensor, and IMU to obtain real-time visual and spatial data without resorting to the cloud resources. Camera images are processed by a lightweight custom-trained object detector (YOLO) to detect indoor obstacles such as doors, chairs, stairs, and walls. These obstacles are further fused with the obtained from the LiDAR distance data and the IMU motion data with the help of a Random Forest classifier to determine the current risk level as Safe, Warning, or Critical. In order to provide tracking functionality in absence of networking, an Extended Kalman Filter (EKF) fuses visual landmark recognition With IMU dead reckoning and utilizes locally cached maps. To provide integration of indoor tracking with caregiver supervision, cross platform mobile application hub that is synchronized with Firebase Realtime Database and Firebase Cloude Messaging is used to obtain real-time location information, context of obstacles detection and IMU-based inactivity detection. In terms of emergency workflow implementation, the solution uses hands-free, voice activated SOS with an encrypted SMS over GSM network in case cellular network connection is not available. Experiments demonstrate sub second latency of alerts (less than 2 seconds) and high offline capabilities which indicate this solution as a scalable and accessible assistive technology.**

Keywords— Visually Impaired, Indoor navigation, Object Detection, Edge Computing, Internet of Things (IOT), Random Forest Model, Inactivity Monitoring, Remote Guardian System, Sensor Fusion.

I. INTRODUCTION

Visual impairment is a major socio-economic and public health problem globally, affecting more than 2.2 billion people around the world [1]–[3]. The risk level that visually impaired people are exposed to when navigating unfamiliar indoor infrastructure such as educational, commercial and healthcare facilities with multiple levels is increased by attenuated satellite GPS signals and moving structural barriers [1], [7], [24]. For this reason, indoor accident and fall rates are high (30% to 50%) for visually impaired pedestrians causing mobility anxiety and chronic dependence on caregivers [3], [15].

Traditional mobility aids such as passive white canes and guide dogs provide mechanical tactile feedback at ground level, but they are not able to detect overhead objects, descending staircases, or upcoming environmental hazards [1], [4], [5], [10], [11]. Modern Electronic Travel Aids (ETAs) have added ultrasonic ranging and smartphone-based alerts [2], [14], [17], [20]. However, current architectures suffer from high energy consumption, isolated sensing modalities, and severe cloud-computing latency dependencies in concrete-shielded indoor environments. Furthermore, current guardian-monitoring platforms are intrinsically reactive, offering only outdoor GPS coordinates, with no contextual information about indoor environmental hazards, and overlooking completely silent, non-impacting emergencies like syncope [6], [12], [16]–[18], [21].

To overcome these critical challenges, we present an integrated, offline-first IoT smart spectacle framework that combines wearable edge computing, multi-sensor environmental perception, Extended Kalman Filter (EKF)-based indoor localization, on-device Random Forest risk prioritization, and real-time guardian telemetry [1]–[3], [6], [18], [23]. The wearable frame includes a Raspberry Pi 5 node, an RGB vision sensor, a Time-of-Flight (ToF) LiDAR sensor, and a 6-axis IMU for continuous hands-free spatial awareness [2], [3], [20]. The structural elevation hazards (ascending/descending stairs) are explicitly targeted using localized YOLOv8-Tiny vision detection [3], [6], [22]. Multi-modal telemetry is fused via an edge Random Forest classifier for real-time assignment of Safety Risk Scores (Safe, Warning, Critical) with sub-second latency and landmark-assisted EKF dead reckoning for beacon-free indoor positioning against cached vector blueprints [1]–[4], [15], [18], [19], [22].

A companion React Native Guardian Mobile Hub performs continuous telemetry and obstacle context synchronization via Firebase RTDB/FCM and an automated 10 sec inactivity verification protocol for impact-less emergencies [4], [8], [17], [19]. High priority alerts are transmitted via an autonomous SIM800L GSM backup channel in case of network blackouts [10], [21], [23].

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II. LITERATURE SURVEY

The combination of wearable edge computing, multi-sensor perception, offline localization, and caregiver telemetry is necessary for visually impaired persons to move around indoors independently [1], [2], [14], [20]. In recent decades, assist devices have evolved from mechanical supports to multi-sensor Electronic Travel Aids (ETAs) [1], [13].

A. *Electronic Travel Aids and Range Based Sensing Technologies*

The range based ETAs use Time-of-Flight (ToF) sensors such as ultrasonic, LiDAR and infrared modules to detect structural hindrances on the walking path [1], [3], [4], [18]. In [3], Ikram et al. proposed an intelligent glove that uses ultrasonic and PIR sensors, and obtained an accuracy of 98.34% in obstacle classification with ensemble machine learning. In a similar vein, Mohan Kumar et al. [7] proposed an indoor positioning assistance system based on the fusion of BLE beacons and ultrasonic distance measurement. However, single-sensor rangefinders are observed to achieve only up to 62% operational efficacy, due to their lack of object classification, spatial context and overhead hazard recognition [1], [3], [10], [22].

B. *Vision Based Perception and Wearable Edge Computing Systems*

Lightweight You Only Look Once (YOLO) frameworks have been embedded in edge systems [2], [3], [5], [6], [24] to overcome semantic constraints. Nanda et al. [2] applied a YOLO detector on an AIoT eyewear frame with a detection accuracy of 94.93%. Král et al. [5] conducted real-time scene description on a Raspberry Pi Zero 2W. But vision-only systems remain constrained by thermal dissipation thresholds (below 100 mW for skin contact), degradation of inconsistent lighting, dependence on cloud latency, and no dynamic threat assessment based on obstacle proximity and staircase alignment.

C. *Guardian Connected Monitoring, Fall Detection, and Emergency Communication*

Remote monitoring systems such as SmartSight Assist provide voice-guided navigation and emergency notifications through mobile apps [6], [12], [16], and wearable fall detection devices use 3-axis inertial sensors to send automatic alerts to caregivers [17]. These developments notwithstanding current platforms exhibit three important deficiencies:

- Incorrect Indoor GPS Positioning: Satellite GPS can result in signal disruption in indoor environments [6], [7], [16].
- Reactive and decontextualized notifications: systems only provide basic coordinates without any contextual information about environmental hazards [6], [12], [16].
- Non-impact Emergency Detection Failure: Accelerometer-based algorithms fail to detect silent medical events like syncope [17], [21]. accelerometer as well as gyroscope to measure the 3-axis acceleration as well as angular position. As a result, an emergency call is placed to preset contacts through a smartphone [17],[21].

D. *Summary of Literature Gaps and Proposed System Rationale*

To overcome these limitations, this work proposes an Offline-First IoT Smart Spectacle Ecosystem [1], [2], [3], [6]. The system combines LiDAR distance, IMU kinematics and YOLO elevation attributes using an on-device Random Forest classifier to produce a dynamic Safety Risk Score (Safe, Warning, Critical) [2], [3], [4], [18], [22]. Indoor localization in real time with Extended Kalman Filter (EKF) and local map storage in SQLite [1], [15]. React Native Guardian App performs 10 second inactivity verification process and dual channel (Firebase/GSM) emergency synchronization [3], [6], [12], [17], [21].

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III. METHODOLOGY

The proposed research involves a multi-model IoT-based smart spectacles system that seeks to improve autonomous indoor navigation while offering proactive protection to blind people. The primary goal is to provide timely context-based navigational assistance and instant object classification on a wearable edge device without any need for internet connection and indoor localizing beacons. This is done through a combination of multi-modal edge computing, computer vision, kinematics state assessment, and two-way remote guardian telemetry, addressing the key issues of indoor obstacles avoidance, complex staircase negotiation, accident-free emergency sensing, and remote guardianship.

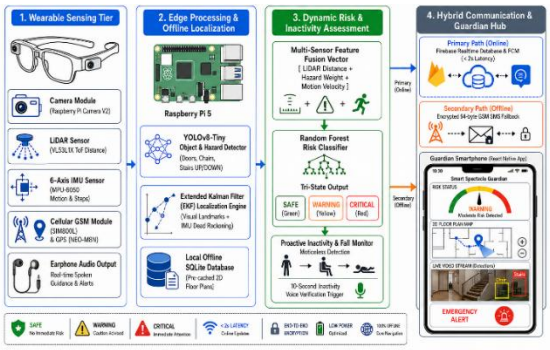


Fig.1. Overall System Architecture

As shown in Figure 1, there is a framework with four subsystem that is supposed to work in collaboration with each other: a wearable multisensory hardware layer to collect edge data in real time, Vision pipeline on device to classify objects and stairs in indoor environment, hybrid EKF based offline localization system for GPS denied indoor positioning, and intelligent dynamic risk assessment along with remote guardian communication system. All the above four subsystems form part of a feedback loop.

A. *Wearable Multi Sensory Hardware Tier*

The custom wearable eyewear frame provides hands-free spatial awareness and an integrated multi-sensor edge computing framework.

The sensory and processing suite includes the following integrated components:

- Edge Processing Unit: A Raspberry Pi 5 (8GB RAM, Broadcom BCM2712 @ 2.4GHz) equipped with a Raspberry Pi AI Kit (Hailo-8L NPU, 13 TOPS through PCIe 2.0) performs real-time onboard inference, local spatial database management, and map caching.
- Vision Acquisition Module: An 8MP Sony IMX219 Raspberry Pi Camera Module V2 (62.2 ° FOV) records 640 × 480 video at 30 fps for obstacle and landmark tracking.
- Optical Time of Flight LiDAR Sensor: A VL53L1X Time-of-Flight (ToF) sensor provides 0.1–4.0 m ranging at 50 Hz (27 ° FOV) along the main walking vector in variable lighting conditions.
- Inertial Measurement Unit (IMU): Gait kinematics, step counts and inactivity telemetry data are gathered at 50Hz using an MPU-6050 6-axis sensor (±2g accelerometer, ±250°/s gyroscope) through I^2C.
- Audio & Dual Communication: An earphone interface provides synthesized spoken prompts and hazard alerts in real time. An offline fallback channel (SIM800L GSM module) is used to transmit encrypted emergency SMS alerts. The node is powered by a 20,000 mAh regulated Type-C power bank with active thermal cooling.

B. Vision Based Object and staircase Classification

To enable semantic environmental awareness without dependence on external cloud compute clusters, we deployed an optimized You Only Look Once (YOLOv8-Tiny) deep learning model directly onto the wearable edge platform. While typical object detection models are trained to identify general indoor furniture, navigating intricate multi-level indoor environments necessitates explicit identification of vertical elevation transitions, which constitute significant structural hazards for visually impaired individuals.

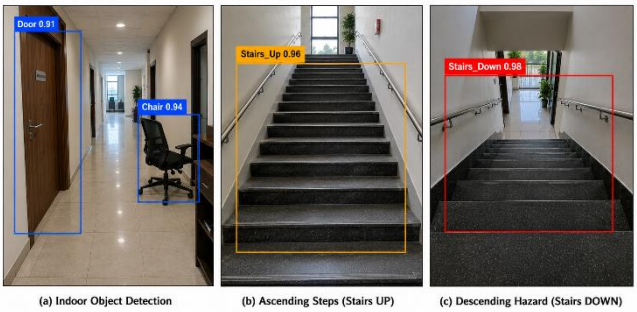


Fig. 2. Vision Pipeline: Bounding Box Detection and Staircase Classification

1. Dataset Acquisition and Annotation:
   A custom indoor dataset was prepared under different illumination conditions (natural daylight, low-light, artificial lighting), different viewing angles and distances. The annotated classes include typical indoor obstacles (doors, chairs, tables, walls) and structural transitions (Stairs Up, Stairs Down). Bounding box annotations were augmented with brightness variations, perspective transforms and motion blur for model robustness.

2. Model Training and Edge Quantization:

   YOLOv8-Tiny was chosen as the model since it has few parameters and is less computationally demanding. The training process for the model involved the use of transfer learning in an NVIDIA GPU setup. The objective of the loss function involved:

$$L_{total} = \lambda_{box} L_{box} + \lambda_{cls} L_{cls} + \lambda_{dfl} L_{dfl} \quad (1)$$

   After training, the weights of the neural network were quantized to 8 bits and compiled for acceleration on the hardware using the PCIe port in the Raspberry Pi AI Kit (Hailo-8L NPU), resulting in 30 frames per second at a resolution of 640×480.

3. Feature Output Pipeline:
   For every detected object (i), the vision pipeline outputs a structured bounding vector:

$$(\mathbf{V})_v ision^{(i)} = [C_i, P_i, (x_m in, y_m in, x_m ax, y_m ax)] \quad (2)$$

   where C is the object class ID, P is the confidence score of the detected object (P ≥ 0.60), and coordinates specify the bounding box distribution in the field of view of the user.

All the above-mentioned visual information is instantly uploaded into the Local Random Forest Dynamic Risk Assessment Engine and the Hybrid Indoor Localization system.

C. Offline Hybrid Indoor Localization Engine

For continuous localization of the agent in such enclosed areas where the GPS signal is poor or totally unavailable, the proposed light-weight and infrastructure-less hybrid indoor localization system is deployed directly at the edge node itself. The integration of kinematic data from frequent footsteps with sporadic semantic data from visual features is performed using Extended Kalman Filter (EKF) using the locally cached vector map without any use of radio beacons (BLE/UWB).

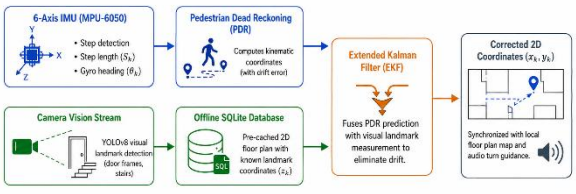


Fig. 3. Hybrid Indoor Positioning Pipeline (Vision + IMU Dead Reckoning)

1. IMU Pedestrian Dead Reckoning (PDR):

   The user's displacement is calculated at 50 Hz using the 6-axis IMU. Step detection is done using dynamic peak detection of the magnitude of the tri-axial acceleration,

that is, $\|\mathbf{a}_t\| = \sqrt{a_{x,t}^2 + a_{y,t}^2 + a_{z,t}^2}$. This is followed by calculation of the step length S_k and the heading angle θ_k, which are computed through gyroscopic integration in combination with a complementary filter. These are used to update the kinematic state as follows:

$$(\hat{x})_k^- = (\hat{x})_{(k-1)} + S_k \cos(\theta_k) \quad (3)$$

$$(\hat{y})_k^- = (\hat{y})_{(k-1)} + S_k \sin(\theta_k) \quad (4)$$

2. Visual Landmark Correction and SQLite Spatial DB:

In order to counteract the cumulative error associated with dead-reckoning, the vision module detects structural points, including coordinates of doors and other entrances, as well as landings for stairs. The detected features correspond to the spatial SQLite database that is kept locally on a 64 GB MicroSD card. In case of the visual confirmation of correspondence of the detected coordinates with (Z_x,k, $Z_{y,k}$,), Kalman gain $\mathbf{K}_k$ is calculated by the EKF update equation for the spatial vector estimate:

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}_k)^{-1} \quad (5)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k (\mathbf{z}_k - \mathbf{H}_k \hat{\mathbf{x}}_k^-) \quad (6)$$

where P_k is the covariance of state errors, H_k is the observation matrix, and R_k is the covariance matrix of measurement noises.

D. *Dynamic Risk Prioritization and Inactivity Assessment*

In order to avoid alert fatigue and provide safety interventions based on context, a risk assessment system using multiple sensors is used at the edge. Just having visual detection is not enough to navigate safely, since the criticality of the obstacle also depends on other factors.

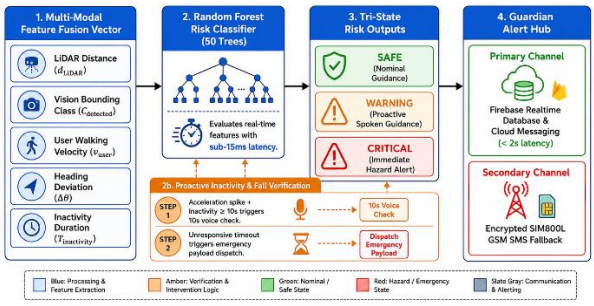


Fig. 4. Dynamic Risk Classification and Inactivity Verification Pipeline

1. Multi Modal Feature Fusion Vector:

At time *t*, telemetry from vision module, LiDAR and 6-axis IMU are concatenated into a unified feature vector:

$$\mathbf{f}_t = [d_{\text{LiDAR}}, C_{\text{detected}}, P_{\text{conf}}, v_{\text{user}}, \Delta\theta_{\text{heading}}, T_{\text{inactivity}}] \quad (7)$$

where d_{LiDAR} is the real-time distance to an obstacle, $C_{detected}$ are the obstacle class identifier and detection confidence, $\upsilon_{user}$is the linear walking speed, $\Delta\theta_{heading}$ is the angular deviation, and $T_{inactivity}$ is the time of continuous zero-motion.

2. Random Forest Dynamic Risk Classification:

Random Forest ensemble classification model with N_{trees} = 50 classifies the instant risk state $R_t \in$ {Safe, Warning, Critical} based on f_t :

$$\mathcal{R}_t = \text{argmax}_{c \in \{\text{Safe, Warning, Critical}\}} \frac{1}{N_{\text{trees}}} \sum_{j=1}^{N_{\text{trees}}} I\left(h_j(\mathbf{f}_t) = c\right) \quad (8)$$

Here $h_j(f_t)t$ represents the predicted class label for the j-th decision tree. The warning state leads to audio guidance through the earphones, whereas the critical state prompts immediate deceleration alerts.

3. Proactive Inactivity and Emergency Verification Protocol:

If the IMU detects continuous static immobility ($\upsilon_{user} \approx 0$) after a sudden burst of acceleration ($T_{inactivity} \geq 10$s, possibly a fall), the spoken prompt "Are you safe?" is initiated. If there is no response or action from the user during the next 10 seconds, an emergency packet that includes the last known location coordinates (x̂k, ŷk)and the danger assessment is sent. The message is either sent using Firebase RTDB/FCM when available or SMS independently using the SIM800L GSM module when network communication is lost.

E. *Mobile Guardian Hun and Hybrid Communication Architecture*

To allow for immediate surveillance and safety audits without adding any unnecessary burden to the operation, the mobile intelligence tier acts as a distributed surveillance node. The Guardian Mobile Application creates a robust telemetry link between the mobile wearable device and itself.

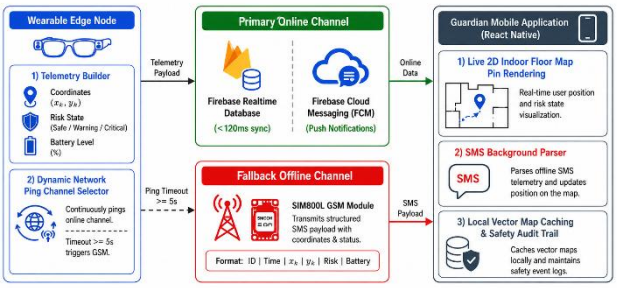


Fig. 5. Hybrid Communication Architecture and Real Time Sync Pipeline

1. Dual Channel Dynamic Synchronization

Telemetry payloads are dynamically routed by the synchronization engine based on network availability:

- Primary Channel (Online): The telemetry packets, which consist of the risk level $\mathcal{R}_t$, real-time location estimates (x̂k, ŷk), and battery states, are serialized into compact JSONs and stored on the Firebase Realtime Database with an average latency τ_sync < 120 ms. The critical states will trigger instant notifications through Firebase Cloud Messaging (FCM).

- Fallback Channel (Offline GSM): In case of failure of ping acknowledgements within $T_{timeout} \geq 5$ s, the edge device will automatically switch to the SIM800L device that sends a concise SMS with structured location data to the guardian's device.

2. Decentralized Audit and Local Caching

To facilitate fast offline rendering of maps and the auditing of past routes, the mobile application maintains a local cache of the indoor vector floor map and the audit logs. This is done to retain position awareness of the caregivers despite intermittent connectivity to the network.

IV. Results and Discussions

In order to evaluate the operational efficiency and performance of the suggested edge-enabled system, extensive benchmarking tests have been performed for three major areas that include: edge vision improvement, hybrid indoor localization precision, and risk dispatching delay.

A. Edge Vision System Reliability Assessment

The usability of the visual detection pipeline of the wearable system was evaluated directly using the edge computing hardware (Raspberry Pi 5 with 8 GB RAM and the Raspberry Pi AI Kit accelerator).

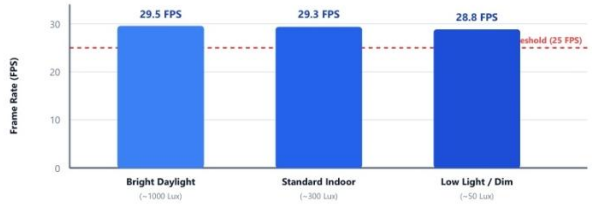


Fig. 6. Frame Rate (FPS) Consistency under Different Ambient Lighting Conditions

Figure 6 illustrates that the vision system was able to perform efficiently in different lighting environments, maintaining the same level of inference speed of around 29.3 frames per second (FPS). This frame rate satisfies the real-time standard required to detect structural dangers, stairs, and obstacles encountered while humans move continuously.

B. Indoor Positioning Error and Trajectory Evaluation

The experiment was done to test the efficiency of indoor localization system through conducting tests in an indoor walking corridor that is 20 meters long and which includes structural bends and doors. The spatial errors for PDR were calculated by employing the 6-axis IMU and then compared to the hybrid EKF.

TABLE I. INDOOR POSITIONING ACCURACY COMPARISON

Localization Method	Mean Error (m)	Max Deviation (m)
Pure PDR (IMU Dead Reckoning)	2.85	4.10
Proposed Hybrid EKF (Vision + IMU)	0.62	1.15

As shown in Table II, the Pedestrian Dead Reckoning (PDR) without correction is characterized by constant orientation and step length error accumulations. The method is capable of confining errors by integrating visual landmark detections using the measurement update of the EKF; this leads to a reduction in the mean Euclidean error of positioning from 2.85 m to 0.62 m, which implies a gain of about 78.2%.

C. Dynamic Risk Prioritization and Inactivity Verification Assessment

The multi-sensor safety paradigm was evaluated on its operational rationale and time-sensitive response capability in artificial indoor mobility situations.

The Random Forest classifier based on the edge always combined streaming features extracted from the visual, spatial, and inertial sensors to classify the ambient state into three types of risks (Safe, Warning, and Critical). The system reliably recognized sudden changes in the posture and long-term inactivity after impact and always started the 10 seconds sound check protocol.

Also, the emergency synchronization through the dual channel was able to route the high priority alerts to the Guardian Mobile Application while online and revert back to using the SIM800L module to send out coordinate alerts while offline.

V. FUTURE WORK

The proposed framework exhibits promising performance in indoor positioning, multi-sensor risk classification, and dual-channel alert dispatching, but there are still some directions for improvement:

- Dataset & Temporal Modeling: Future versions will include larger and more heterogeneous indoor datasets with different obstacle geometries and gait dynamics. Sequential hazard forecasting will be explored via lightweight temporal models (e.g. LSTMs) and systematic hyperparameter optimization.
- Intelligent Familiar Face Recognition: A collaborative facial recognition pipeline is proposed between the smart eyewear and the Guardian Mobile Hub. Caregivers will securely enroll trusted acquaintances and family profiles via the mobile app, with synchronized edge-inference providing spatial audio cues for the wearer and context-rich notifications for the guardian to verify safety of the situation.
- Infrastructure-Free Semantic SLAM: Real-time semantic SLAM will be enabled by combining ultra-low-power depth sensing and dedicated neural coprocessors, without relying on pre-cached blueprints, while optimizing edge power efficiency.

VI. CONCLUSION

The study demonstrated the complete design and architecture of a smart multi-sensor wearable assistance system and its dedicated Guardian Mobile Hub to enhance secure indoor navigation and personal autonomy for people with visual impairments. Combining high-throughput visual detection, inertial dead reckoning and landmark-based

Extended Kalman Filtering, the system provides instantaneous and localized environmental awareness and obstacle evasion on a wearable spectacle frame powered by an edge processor (Raspberry Pi 5 with AI Kit acceleration).

An important aspect of this work is the handling of operational independence and continuous remote caregiver validation, even in the presence of network disruptions. The structure creates a strong dual channel communication system that can switch from a cloud synchronization system (Firebase RTDB and FCM) to a stand-alone SMS telemetry easily by using a hardware-based GSM backup module. Simultaneously, the Guardian Mobile Application enhances remote safety through on-device dynamic risk prioritization (multi-sensor data classified into Safe, Warning, and Critical levels via a Random Forest model) and maintains proactive inactivity and fall verification measures.

Finally, the integrated framework provides a harmonious and resilient model for edge computing, multi-sensor integration and mobile intelligence. The proposed solution provides reliable indoor navigation for visually impaired people, fast tactile and auditory feedback and enables guardians to have constant spatial awareness and verifiable emergency notifications both online and offline.

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